Main Design Improvements

- As there were some issues with efficiency and thermal management of the bricks, two components were studied to be replaced
  - The original power MOSFETs (STB57N65M5) caused high switching losses and was replaced by SIHFS9N60A
    - The new MOSFET caused a significant increase in the efficiency of the bricks (from ~58% to ~72%)
    - The new MOSFET had a 65% smaller gate-source capacitance
  - The output inductor was getting a bit hot, and was replaced with an inductor with smaller series resistance
- The improvement in the efficiency helped with the limitations of the cooling system
  - A brick efficiency of at least 65% was required for the cooling plant with 77 kW capacity
- Due to limitation in cabling from the service room to the modules, the number of signals for controlling each brick had to be reduced

Design	Signals	Number of Wires
Legacy	individual enable signals + returns common start-up signal	18
Phase-2	individual tri-state signals (turn-on, turn-off, keep state) + common return	9